

Summary of the interviews

2025-04-08

Results

Scoring - overview

The questionnaires originated more than 500 scores. About half of the responses (266) were including a clear and unequivocal quantitative score. A large number (123) were not considered relevant by interviewee. Answers that were considered relevant but did not provide a clear score were 150, among which the large majority (105) contained enough information to attempt an interpretation.

Table S1: test

Score	Frequency
x	266
-999	123
999	105
-998	35
-997	5

Scoring - does not apply (-999)

For each interviewee and event we computed the proportion of the answers scored as -999. These events with more than 50% of answers scored as -999 were excluded. The combination of events/styles excluded are the reported in the table. The image reports the proportion of answers indicating that the event is relevant.

desc	style
Heatwave in the winter season	coastal
Heatwave in the winter season	coastal
Severe icing (in the winter season)	coastal
Severe icing (in the winter season)	subsistence
Severe icing (in the winter season)	coastal
Algal bloom (in the summer season)	coastal

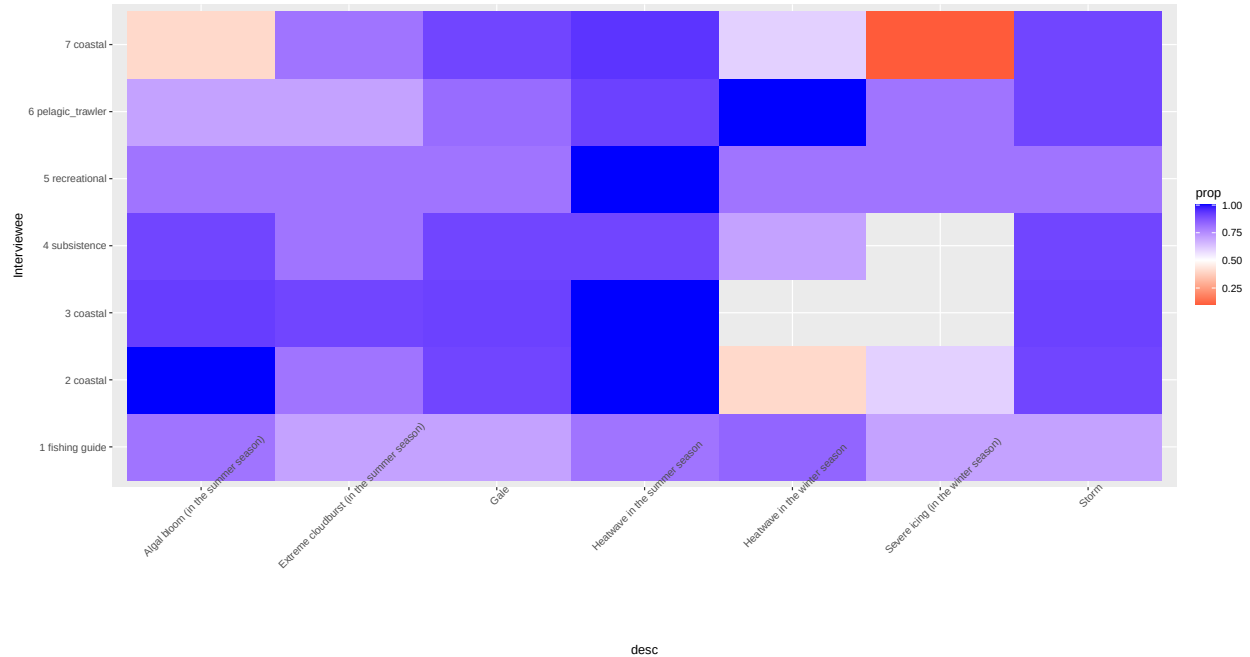


Figure S1: Proportion of responses that were considered relevant by interviewee and event

Scoring - Garbled (-997)

These answers were clearly not matching the question. We attempted a reconstruction as described in next phases. Most of these relates to the catch composition: we were interested in the behavior of the fish, not of the fisherman. Anyway we can use this information to support other aspects (e.g.: how easy is to avoid an event).

style	description	text
fishing guide	Do you catch a different composition of species during such extreme weather events?	: 5 sea trout on open water, instead of perch and pike under the ice
fishing guide	Do you catch a different composition of species during such extreme weather events?	: 5 redirect the fishing down below the thermocline
fishing guide	Do you catch a different composition of species during such extreme weather events?	More about changing location (deeper), 5 choose other species (we fish pike & perch, sea trout)
fishing guide	Do you catch a different composition of species during such extreme weather events?	: 2 sometimes; if it blows too hard and we are supposed to fish trout, then sometimes we have to go in and fish perch & pike instead
fishing guide	How much do you think the answers to these questions (questions 2-7) vary between fishers in the group you represent? [1-5; from 1 – not at all, to 5 – very much]	5, since I am chairman I have the mandate to represent the others

recreational	Do you experience that the catchability of the fish changes due to such extreme weather events?	If there's no ice, you can't catch them with an ice rod, and you don't put in a boat just for a short ice-free spell.
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Interpretation - Expert scoring

Expert scoring of keywords produced the following results, which were used to assign scores to 87 answers:

question	coding	answer	AG	ENA	JO	mean	sd
is it possible [to avoid something]	1-5; from 1 – very difficult, to 5 – very easy	hard	2	2	2	2	0.0
is it possible [to avoid something]	1-5; from 1 – very difficult, to 5 – very easy	easier	3	4	4	4	0.6
is it possible [to avoid something]	1-5; from 1 – very difficult, to 5 – very easy	to some extent	3	4	3	3	0.6
is it possible [to avoid something]	1-5; from 1 – very difficult, to 5 – very easy	no	1	1	1	1	0.0
is it possible [to avoid something]	1-5; from 1 – very difficult, to 5 – very easy	not difficult	4	3	5	4	1.0
is it possible [to avoid something]	1-5; from 1 – very difficult, to 5 – very easy	it is possible	3	4	4	4	0.6
is it possible [to avoid something]	1-5; from 1 – very difficult, to 5 – very easy	probably not entirely impossible	2	2	2	2	0.0
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	you can't [do something]	1	1	1	1	0.0
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	yes but [in a different way]	3	3	3	3	0.0
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	it has a big negative impact	1	1	2	1	0.6
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	still possible [to do something]	3	3	3	3	0.0
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	most do not [do something]	no info	2	2	2	0.0
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	impossible [to do something]	1	1	1	1	0.0
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	nothing can be done	1	1	1	1	0.0

is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	we can do little	2 or 4, depending on Swedish wording (we can do little vs we can do a little)	2	3	2	0.7
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	I would try to	no info	3	4	4	0.7
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	we wait	no info	1	??	1	NA
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	sometimes	3	3	3	3	0.0
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	i'll do it anyway	no info	4	4	4	0.0
is it [something] different when [event] happens	1-5; from 1 – very similar, to 5 – very different	very similar	1	1	1	1	0.0
is it [something] different when [event] happens	1-5; from 1 – very similar, to 5 – very different	small difference	2	2	2	2	0.0
is it [something] different when [event] happens	1-5; from 1 – very similar, to 5 – very different	it may be different	4	4	3	4	0.6
is it [something] different when [event] happens	1-5; from 1 – very similar, to 5 – very different	it has an effect	no info on direction	3	4	4	0.7
is it [something] different when [event] happens	1-5; from 1 – very similar, to 5 – very different	it affects	no info on direction	3	4	4	0.7
does [event] makes [something] worse or better	1-5; from 1 – much worse, 3 – no difference, to 5 – much better	it is harder to do	2	2	2	2	0.0
does [event] makes [something] worse or better	1-5; from 1 – much worse, 3 – no difference, to 5 – much better	there is a marginal negative effect	2	2	2	2	0.0
does [event] makes [something] worse or better	1-5; from 1 – much worse, 3 – no difference, to 5 – much better	it is more difficult	2	2	2	2	0.0
does [event] makes [something] worse or better	1-5; from 1 – much worse, 3 – no difference, to 5 – much better	it get worse	2	2	2	2	0.0
does [event] makes [something] worse or better	1-5; from 1 – much worse, 3 – no difference, to 5 – much better	there is a little negative impact	2	2	2	2	0.0
does [event] makes [something] worse or better	1-5; from 1 – much worse, 3 – no difference, to 5 – much better	not so big impact	3	2	3	3	0.6

does [event] makes [something] worse or better	1-5; from 1 – much worse, 3 – no difference, to 5 – much better	a little worrisome	no info	2	2	2	0.0
does [event] makes [something] worse or better	1-5; from 1 – much worse, 3 – no difference, to 5 – much better	it certainly has a negative effect	2	1	1	1	0.6
does [event] makes [something] worse or better	1-5; from 1 – much worse, 3 – no difference, to 5 – much better	is good	4	4	4	4	0.0
how much is [something] important	1-5; from 1 – not important at all, to 5 – very important	fairly important	3	3	3	3	0.0
how much is [something] important	1-5; from 1 – not important at all, to 5 – very important	not a strong link	2	2	2	2	0.0
how much is [something] important	1-5; from 1 – not important at all, to 5 – very important	it has a real significance	5	4	5	5	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	it has great effect	5	4	5	5	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	it is big effect	5	4	5	5	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	it may have an effect	3	2	3	3	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	it has big impact	5	4	5	5	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	Quite small	2	2	2	2	0.0
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	not so much	2	2	2	2	0.0
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	not really	1	1	1	1	0.0
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	it does	3	2	3	3	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	quite a lot	4	4	4	4	0.0
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	some	2	3	3	3	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	some, but less than other	2	2	2	2	0.0
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	greatly	4	5	4	4	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	most do not	2	2	2	2	0.0

does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	no	1	1	1	1	0.0
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	improve	4	3	4	4	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	a lot	4	5	5	5	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	it's strong	5	5	4	5	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	pretty much	4	4	4	4	0.0
How much concern for your personal safety	0-2; 0 – no concern, 1 – personal injuries, 2 – life-threatening	big concern	no info bc difference between 1 & 2 is type of concern, not level alone	2	2	2	0.0
How much concern for your personal safety	0-2; 0 – no concern, 1 – personal injuries, 2 – life-threatening	some risk	no info bc difference between 1 & 2 is type of concern, not level alone	1	1	1	0.0
How much concern for your personal safety	0-2; 0 – no concern, 1 – personal injuries, 2 – life-threatening	worrisome	no info bc difference between 1 & 2 is type of concern, not level alone	1	1	1	0.0
does [something] allows you to change location	0-2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible	sometimes, but not always	1	1	1	1	0.0
does [something] allows you to change location	0-2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible	it allows a lot of freedom	2	2	2	2	0.0

Interpretation - AI scoring

This option was applied to a limited number of responses.

Q1

Question: How much do you think they vary among the group of people you represent? Possible answers: [1–5; from 1 – not at all, to 5 – very much]

Observed Answer: it is so personal, some do not care at all, they just drink beer all day; we discuss things like this all the time, most are probably quite stressed by such situations

Score: 4

Why: The reply indicates clear heterogeneity — a minority who “do not care at all” vs a majority who are “quite stressed.” That is meaningful variation but phrased so that most are similar (stressed) with a minority different. In your training examples, minority exceptions vs majority agreement tended to map to a high-but-not-maximum value. Confidence: medium–high.

Q2

Question: How much do you think they vary among the group of people you represent? Possible answers: [1–5; from 1 – not at all, to 5 – very much]

Observed Answer: There are some who have rights but don’t practice at all, and then some who practice like I do, just now and then. Then there are those who have leased out their permits, and in those cases, there is somebody who practices quite intensively

Score: 5

Why: The answer describes the full spectrum (no practice → occasional → very intensive). That explicit broad distribution matches your examples that consistently received top-end variability scores. Confidence: high. Q2 Question: How likely is it that you explore new techniques or strategies? Possible answers: [1–5; from 1 – very unlikely, to 5 – very likely] *Observed Answer:* Would probably try to find solutions; activity would certainly change, and then people would follow that change; there is probably some variation among people depending on what one is specialized in *Score:* 4 *Why:* The speaker says they “would probably try” and expects change and follow-up by others — language that in your dataset equates to a positive likelihood but not the absolute “definitely” (which you typically scored a 5). The note about variation reduces absolute certainty slightly. Confidence: medium–high.

Q3

Question: If your income from activity A decreases, can you complement or replace it? Possible answers: [1–5; from 1 – very unlikely, to 5 – very likely]

Observed Answer: It would be very hard to replace activity A it with other income. Most have other jobs in winter, but A is such a significant part of income

Score: 1

Why: The phrase “very hard to replace” and “A is such a significant part” matches wording in your dataset that mapped to the low end (very unlikely) — even though there are seasonal other jobs, the respondent explicitly indicates replacement is essentially not feasible. Confidence: high.

Q4

Question: How easy is it to avoid an extreme weather event by going to another location?

Possible answers: [1–5; from 1 – very difficult to avoid, to 5 – very easy to avoid] *Observed Answer:* I Stop my activity *Score:* 1 *Why:* The respondent says they stop the activity instead of relocating to avoid the event — that implies relocation is not used/feasible. In your training set, answers that indicate “we don’t/stop” were scored at the difficult end (1). Confidence: medium — short reply is slightly ambiguous about whether “stop” means temporarily pause or that relocation is impossible, but both interpretations point to very low ease of avoiding by moving.

Q5 Question: If the following type of extreme event occurs, how likely is it that you go out and do the activity? Possible answers: [1–5; from 1 – very unlikely, to 5 – very likely] *Observed Answer:* we wait because the product can spoil *Score:* 1 *Why:* “We wait” because of spoilage risk indicates they do not go out during the event — i.e., very unlikely. In your dataset, similar phrasing (“we wait / won’t go because...”) mapped to the lowest likelihood. Confidence: high.

Interpretation - left unscored

These were left unscored either because are mainly needed to create narrative or because their implementation would require further revision of the network. For example there is a lot of mentions to wind direction, which is not yet implemented.

style	description	text
coastal	Do current management regulations make it possible to choose other fishing locations? Briefly describe what possibly limits your choice.	Generally, we have quite a lot of mobility in terms of management; fixed gear locations are accessible fishing spots. Only for salmon fishing the license is tied to a specific boat, so you can’t switch to a larger boat. For net fishing it doesn’t matter. (Example given: switching to a larger and safer boat than the one tied to the salmon license in order to haul the salmon trap during gale, but then getting a penalty on the license).
coastal	Do you experience that the catchability of the fish changes due to such extreme weather events?	Yes – after a gale from the right direction, herring can blow in, salmon may swim more too (E, NE in summer). =4. From W & NW it gets worse instead.
coastal	Do you experience that the catchability of the fish changes due to such extreme weather events?	Yes – after a gale from the right direction, herring can blow in, salmon may swim more too (E, NE in summer). =4. From W & NW it gets worse instead.
coastal	Do you experience that the catchability of the fish changes due to such extreme weather events?	Same, just stronger wind.
coastal	Do you experience that the catchability of the fish changes due to such extreme weather events?	Same, just stronger wind.
coastal	How much damage can the extreme weather event cause to your gear and boat? Explain! several answers are possible]	Could damage or destroy gear/boat, or cause minor damage
coastal	How much damage can the extreme weather event cause to your gear and boat? Explain! several answers are possible]	Could damage or destroy gear/boat, or cause minor damage
coastal	Do you have gear, equipment and boat that allow you to choose other fishing locations (to avoid an extreme weather event)?	Yes, can move around for herring, whitefish, perch. Only salmon gear is fixed. Our operations use active gear, so we can relocate, but only short distances – catch must be landed quickly. (Older generations even traveled to Trosa/Grisslehamn.)

subsistence	Do current management regulations make it possible to choose other target species? Briefly describe what possibly limits your choice.	Eel fishing ban; voluntary fishing stop Mar 15–May 15 in some spawning bays. We must restore predatory fish stocks. We think fish shelters (we’ve built 10 over 10 years) have limited effect at such low stock levels.
recreational	Do you catch a different composition of species during such extreme weather events?	Not directly, but on the coast with prolonged winds you can get upwelling; perch may move closer inshore, and trout & whitefish may appear shallower than normal. Seen within a season
recreational	Do you catch a different composition of species during such extreme weather events?	Not directly, but on the coast with prolonged winds you can get upwelling; perch may move closer inshore, and trout & whitefish may appear shallower than normal. Seen within a season
recreational	How much does your “community” benefit from your fishing? Explain!	Some drive among those eager to show off successful fishing. But being part of that community doesn’t really require fishing.
coastal	Do current management regulations make it possible to choose other target species? Briefly describe what possibly limits your choice.	There isn’t much else to fish for; we catch other fish species, but in much smaller quantities, such as pike and pikeperch; salmon is subject to quotas; herring too, but we don’t fill the coastal quota, and haven’t done so for several years.

Reconstruction (-998)

For these questions we try to inspect closely related answers to see if we can borrow the scoring. These numbers are stored and then used to train the BN (so are not reported in section XXX)

- “Personal” feelings

Personal feelings answers deal with personal decisions taken in face of an event. This is personal in first place and so any “borrowing” should be done with caution. We can inspect closely related fishing styles. We have only two records that needs to borrow from other in the Human category.

style	description	desc	value
coastal	How much concern for your personal safety do these extreme weather events bring?	Storm	-998
recreational	How easy is it to avoid an extreme weather event by going to another fishing location?	Severe icing (in the winter season)	-998

The coastal fisherman category is represented by three interviewee. We do not need to reconstruct anything at this stage because the BN can be informed by the other respondents. For the recreational fisherman we can try to look for the same answer in the fishing guide. Unfortunately this reflecting too much the commercial purpose of the fishing guide and cannot be applied to the recreational.

description	text	value
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How easy is it to avoid an extreme weather event by going to another fishing location?	Easy for me to switch but ask what the customers want (e.g., Americans who are booked for ice fishing). Large differences from the normal are a problem, with bookings we do not know what we shall book (e.g., closed Nyköpingsån fishing due to heatwave). The customers also have to be flexible. 2 for those that are not dangerous; 1 for what is dangerous (cannot conduct activities)	1
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- metier characteristics

Metier characteristics answers deal with technical aspect of the fishing gear or the gear/fish interaction in face of an event. A similar gear can be used by different categories (e.g.: nets, rods). Also, some gears (salmon traps) are very specific and only some people do mention. We have a handful of missing answers (total = 7).

style	description	desc	value
recreational	Do you experience that the catchability of the fish changes due to such extreme weather events?	Heatwave in the winter season	-997
coastal	Do you experience that the catchability of the fish changes due to such extreme weather events?	Severe icing (in the winter season)	-998
coastal	Do you experience that the catchability of the fish changes due to such extreme weather events?	Algal bloom (in the summer season)	-998
recreational	Do you experience that the catchability of the fish changes due to such extreme weather events?	Algal bloom (in the summer season)	-998
fishing guide	Do you experience that the catchability of the fish changes due to such extreme weather events?	Extreme cloudburst (in the summer season)	-998
fishing guide	Do you experience that the catchability of the fish changes due to such extreme weather events?	Storm	-998
recreational	Do you experience that the catchability of the fish changes due to such extreme weather events?	Storm	-998

The coastal fisherman category is represented by three interviewee. We do not need to reconstruct anything at this stage because the BN can be informed by the other respondents. For the recreational and fishing guide, they can inform each other for the effects of winter heatwaves and cloudburst on catchability.

style	gear	description	desc	text	value
fishing guide	ice fishing	Do you experience that the catchability of the fish changes due to such extreme weather events?	Heatwave in the winter season	1 ice fishing, there is no catchability if there are no fish; however, the fish like when it gets warmer, so if it's spin fishing then 5	1
fishing guide	rod	Do you experience that the catchability of the fish changes due to such extreme weather events?	Heatwave in the winter season	1 ice fishing, there is no catchability if there are no fish; however, the fish like when it gets warmer, so if it's spin fishing then 5	5

recreational	NA	Do you experience that the catchability of the fish changes due to such extreme weather events?	Extreme cloudburst (in the summer season)	No.	3
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- Fish related

The category fish related inform on ecological or physiological characteristics of the fish, which are expected to not be dependent on gear or people. We can try to borrow from other answers dealing with similar species and environment. We have a variety of missing answers falling in this category. The main problems are with species composition (n=X) and fish condition (n=Y). Since this affects the fish, we can inform the missing answers depending on people that attempted a score.

style	description	desc	value
fishing guide	Do you catch a different composition of species during such extreme weather events?	Heatwave in the winter season	-997
subsistence	Do you catch a different composition of species during such extreme weather events?	Heatwave in the winter season	-998
fishing guide	Do you catch a different composition of species during such extreme weather events?	Heatwave in the summer season	-997
subsistence	Do you catch a different composition of species during such extreme weather events?	Heatwave in the summer season	-998
fishing guide	Do you catch a different composition of species during such extreme weather events?	Severe icing (in the winter season)	-998
coastal	Do you catch a different composition of species during such extreme weather events?	Severe icing (in the winter season)	-998
recreational	Do you catch a different composition of species during such extreme weather events?	Severe icing (in the winter season)	-998
fishing guide	Do you catch a different composition of species during such extreme weather events?	Algal bloom (in the summer season)	-997
coastal	Do you catch a different composition of species during such extreme weather events?	Algal bloom (in the summer season)	-998
fishing guide	Do you catch a different composition of species during such extreme weather events?	Extreme cloudburst (in the summer season)	-998
subsistence	Do you catch a different composition of species during such extreme weather events?	Extreme cloudburst (in the summer season)	-998
fishing guide	Do you catch a different composition of species during such extreme weather events?	Gale	-997
subsistence	Do you catch a different composition of species during such extreme weather events?	Gale	-998
coastal	Do you catch a different composition of species during such extreme weather events?	Gale	-998
fishing guide	Do you catch a different composition of species during such extreme weather events?	Storm	-998
coastal	Do you catch a different composition of species during such extreme weather events?	Storm	-998

subsistence	Do you catch a different composition of species during such extreme weather events?	Storm	-998
coastal	Do you catch a different composition of species during such extreme weather events?	Storm	-998
recreational	Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?	Severe icing (in the winter season)	-998
subsistence	Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?	Algal bloom (in the summer season)	-998
coastal	Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?	Extreme cloudburst (in the summer season)	-998
subsistence	Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?	Extreme cloudburst (in the summer season)	-998
coastal	Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?	Gale	-998
subsistence	Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?	Gale	-998
coastal	Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?	Storm	-998
subsistence	Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?	Storm	-998

By inspecting the distribution of answers regarding species captured it emerge consistency, with some more variability observed in summer heatwave (expected). We observe one outlier for algal blooms. This is to be revised.

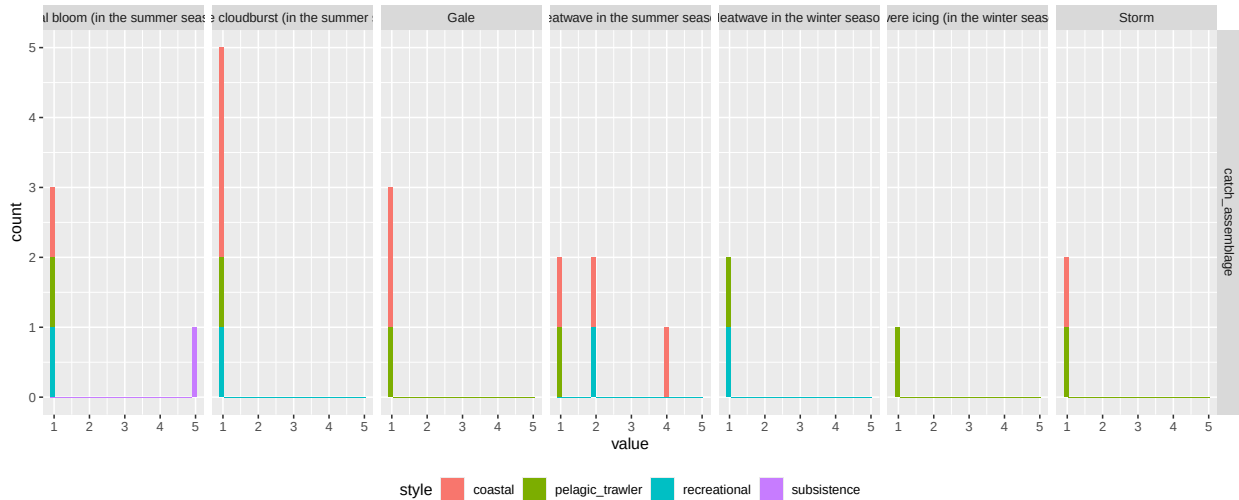


Figure S2: Are the fish in the catch of worse condition or quality [...]?- From 1 much worse to 3 same to 5 much better

By inspecting the distribution of answers regarding fish condition, it emerge a pattern that regards pelagic

trawler. This does not apply to other people (it is damage to fish due to rolling in big waves). Other responses can be used.

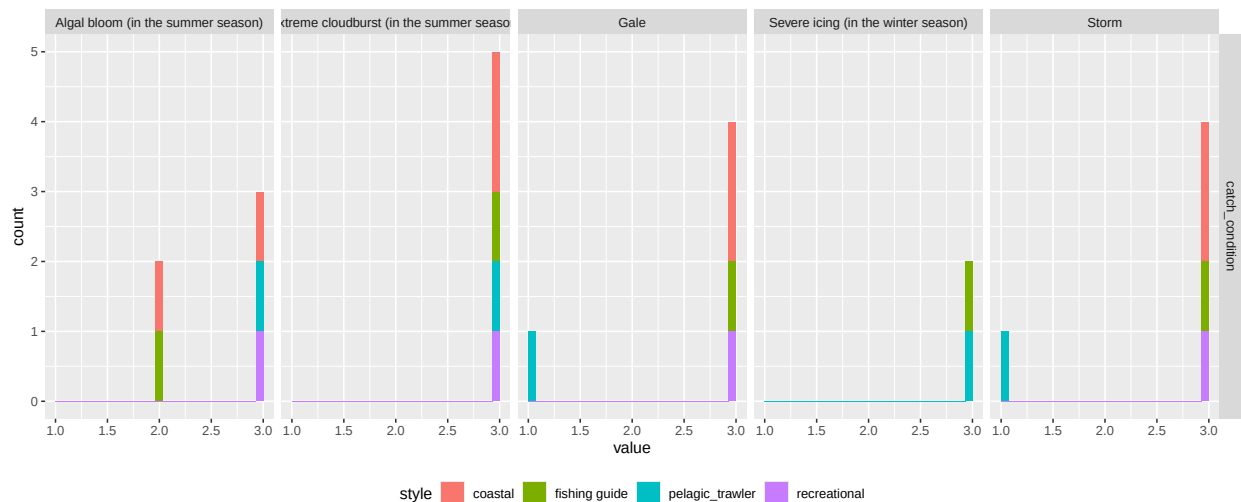


Figure S3: Distribution of answers to the question: Do you catch a different composition of species during such extreme weather events?- From 1 very similar to 5 very different

- Other

What remains are questions regarding uncertainty, long term effects, and change in fishing practices. Long term effects and change in fishing practices are to be used to support narrative and we do not really need them at this stage. It is more important to have some information on uncertainty-related questions.

style	description	desc	value
coastal	Do this extreme weather events have long-term effects on how you fish or catch composition	Severe icing (in the winter season)	-998
coastal	Do this extreme weather events have long-term effects on how you fish or catch composition	Algal bloom (in the summer season)	-998
coastal	Do this extreme weather events have long-term effects on how you fish or catch composition	Extreme cloudburst (in the summer season)	-998
fishing guide	How much do you think the answers to these questions (questions 2–7) vary between fishers in the group you represent? [1–5; from 1 – not at all, to 5 – very much]	NA	-997
recreational	Do you change anything in how you normally fish when such an extreme weather event occurs?	Severe icing (in the winter season)	-998
coastal	Do you change anything in how you normally fish when such an extreme weather event occurs?	Storm	-998
subsistence	How much do you think the answers to the questions about personal well-being (questions 32–34) vary between fishers in the group you represent	NA	-998